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Trademarks
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Contact information
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General Precautions in the Handling of Microprocessing Unit and Microcontroller Unit Products

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1. Overview

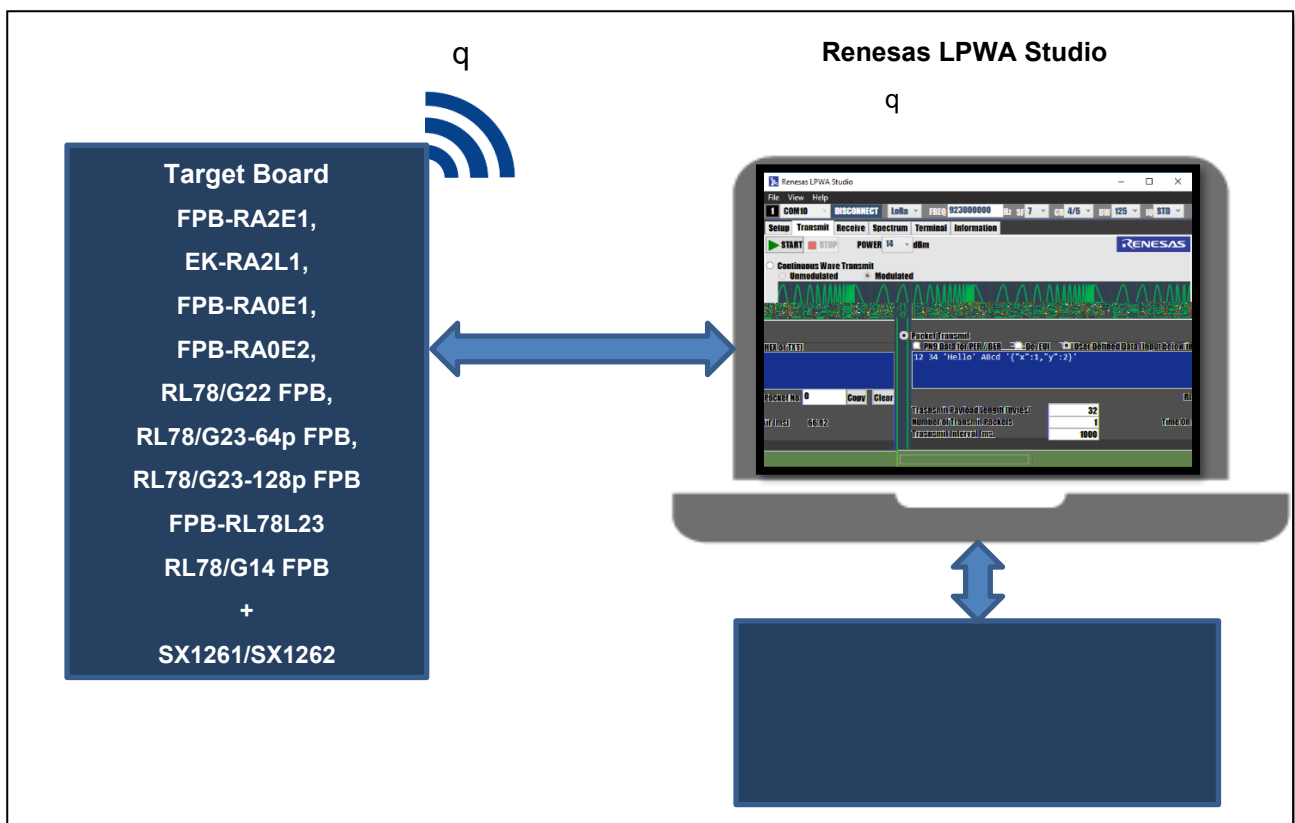
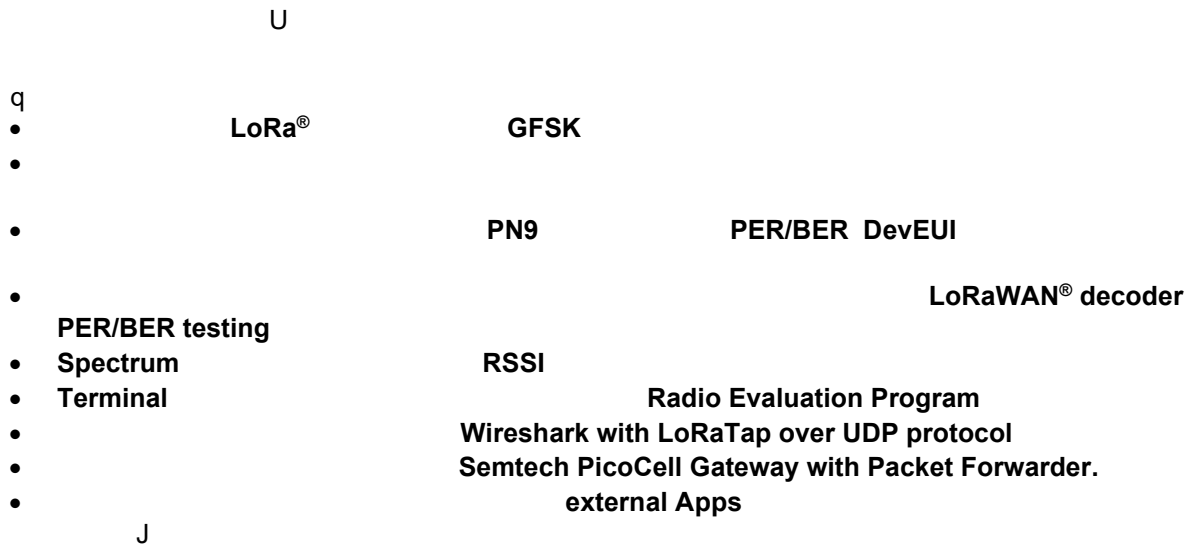


Figure 1. Renesas LPWA Studio System Overview

2. Requirements

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- q
- q

q
q
q

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-

.NET Framework 4.8

RadioEvalApp V4.70 or later

3. Getting Started

3.1 Setup the Radio Evaluation Program

q

- q *RL78/G23, RL78/G22, RL78/L23, RL78/G14 LoRa®-based Wireless Software Package*
- q *RA2E1, RA2L1, RA0E1, RA0E2 LoRa®-based Wireless Software Package*

3.2 Install and Uninstall the Renesas LPWA Studio

IMPORTANT

Install the Renesas LPWA Studio:

Install.

.NET Framework 4.8

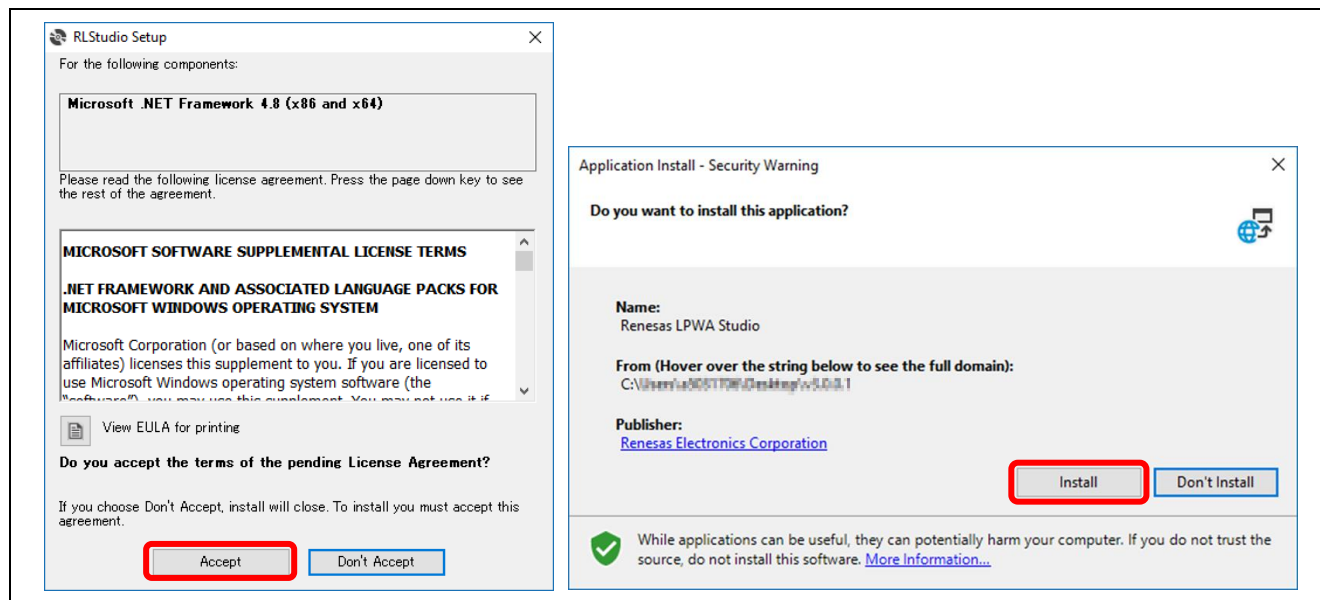


Figure 2. Renesas LPWA Studio Installation

Uninstall the Renesas LPWA Studio:

Start Menu

Settings

Apps

Renesas LPWA Studio

Uninstall

4. Using Renesas LPWA Studio [Basic]

4.1 Connect and Disconnect the Radio Evaluation Program

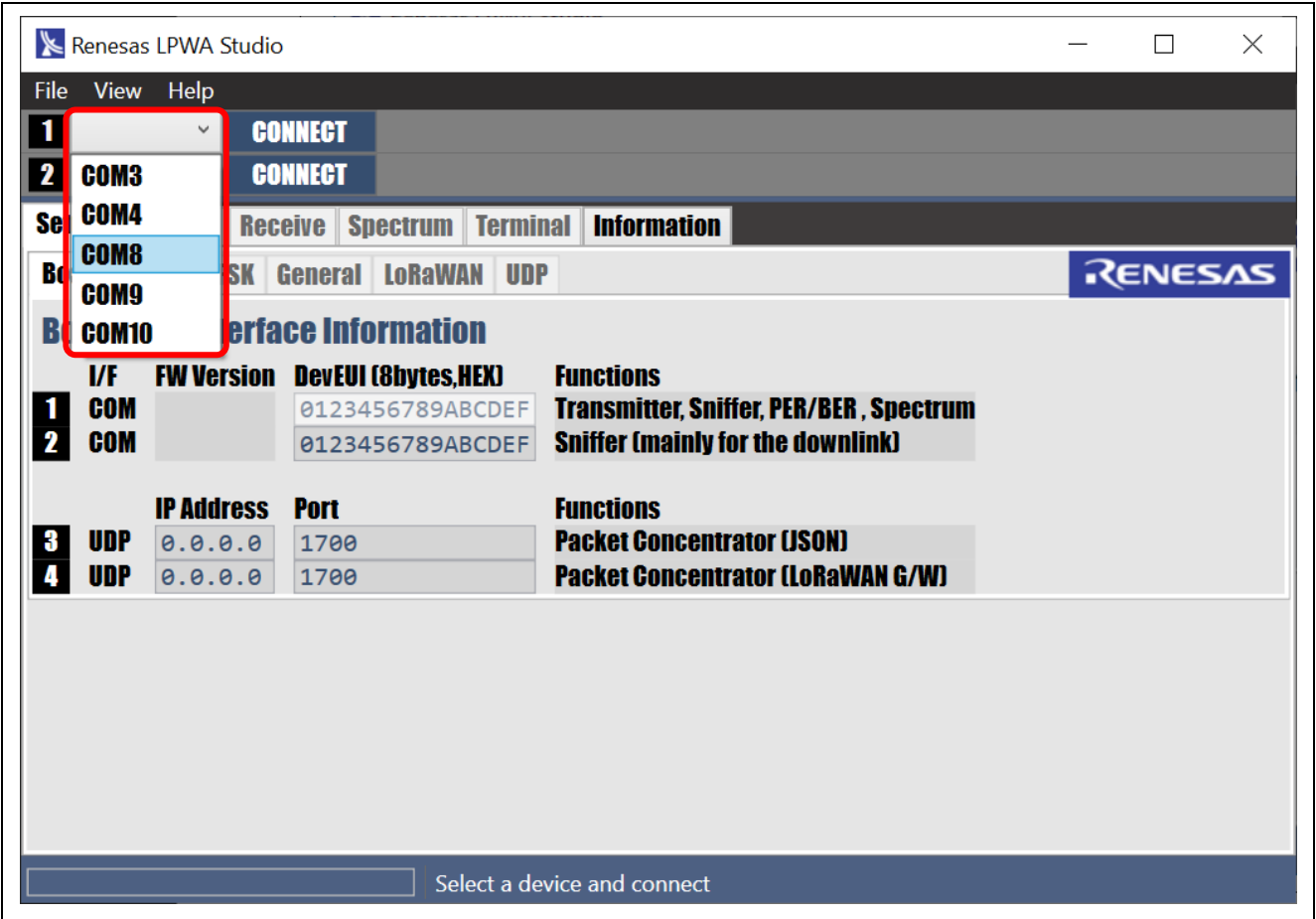


Figure 3. Connecting Radio Installation Program

CONNECT

DISCONNECT

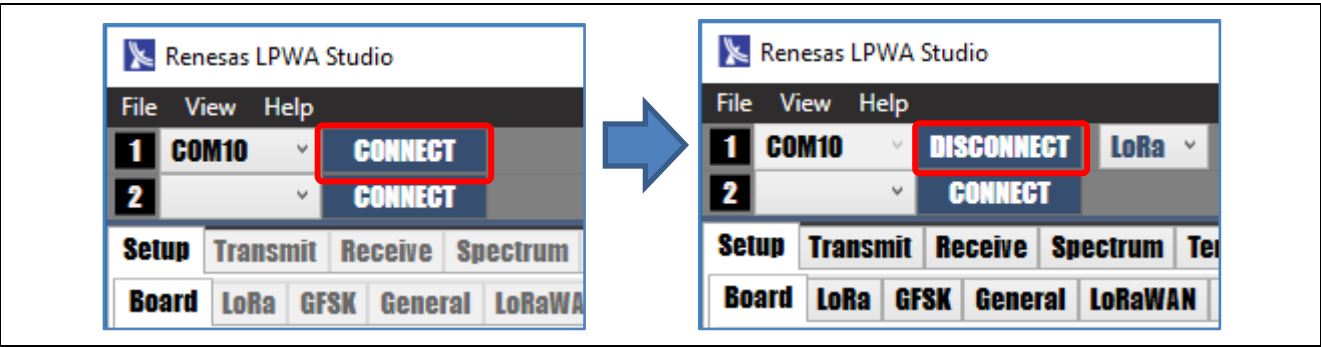


Figure 4. Connection and Disconnection Buttons

DISCONNECT

4.2 Select Modem, Frequency and Radio Parameters

q

q LoRa q SF CR BW IQ

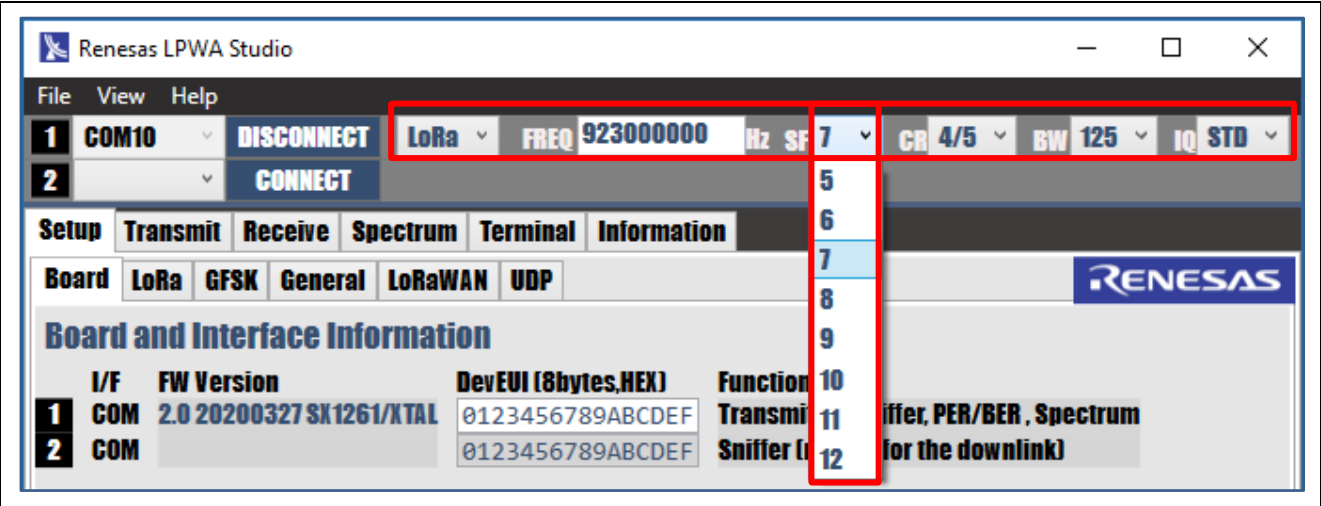


Figure 5. LoRa Parameters

q GFSK DR q Fdev BW

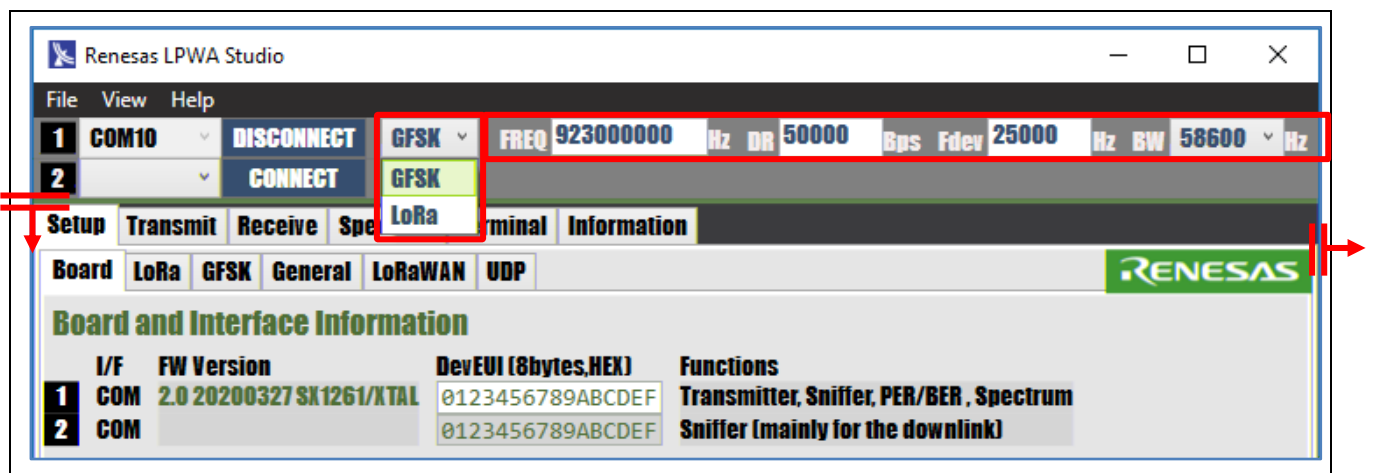


Figure 6. GFSK Parameters

4.3 Configure Packet Parameters (optional)

Setup LoRa GFSK

q

For LoRa

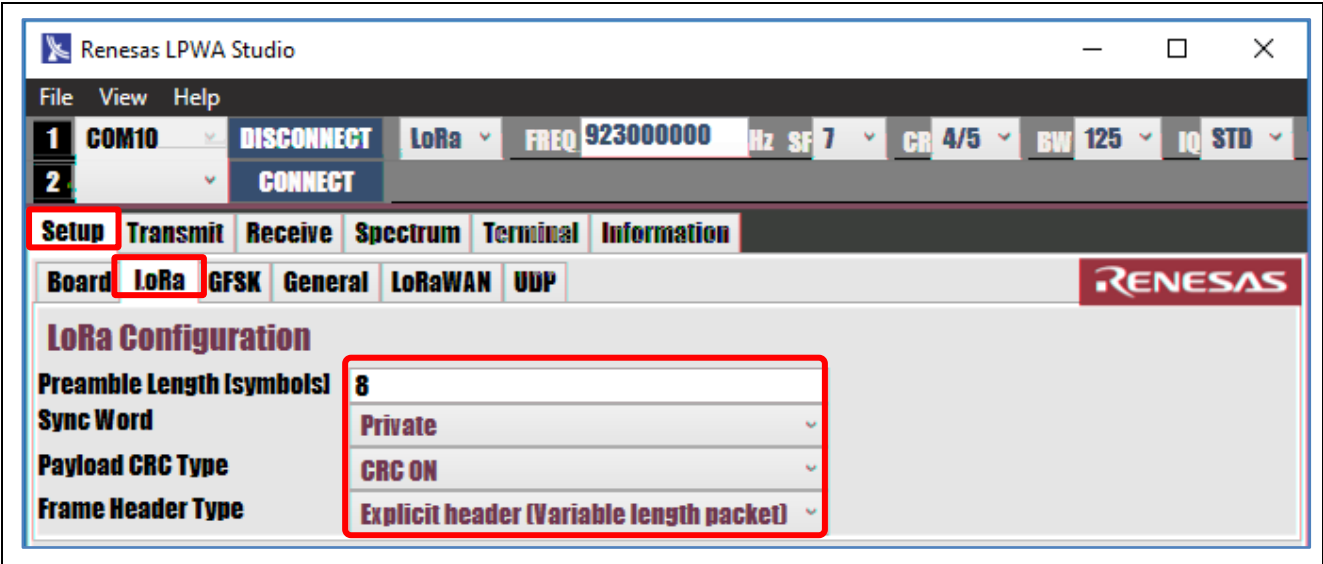


Figure 7. LoRa Packet Parameters Configuration

For GFSK

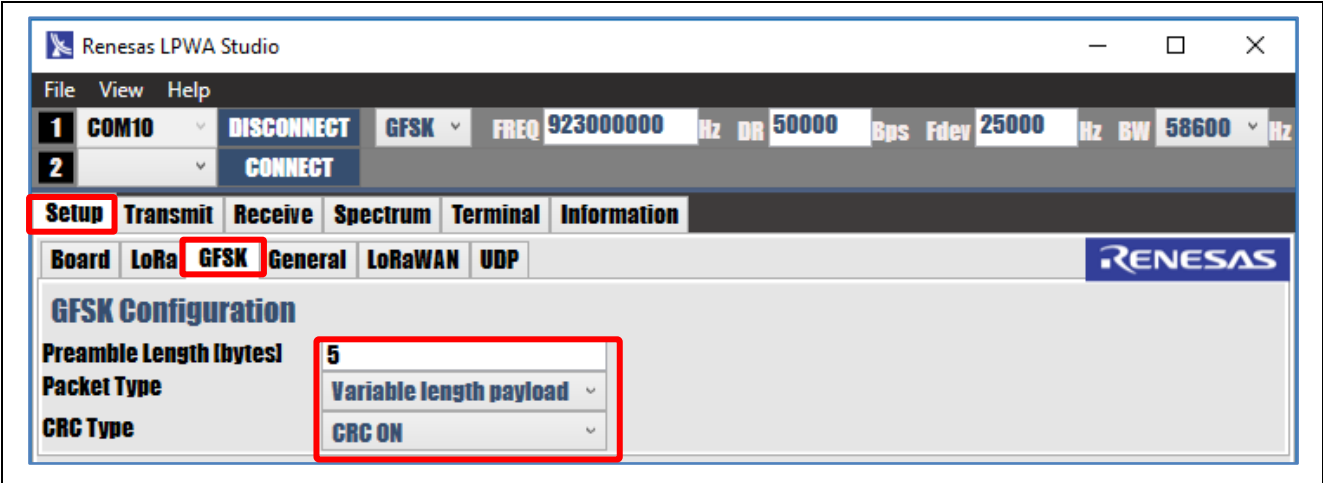


Figure 8. GFSK Packet Parameters Configuration

Boosted Gain

LBT

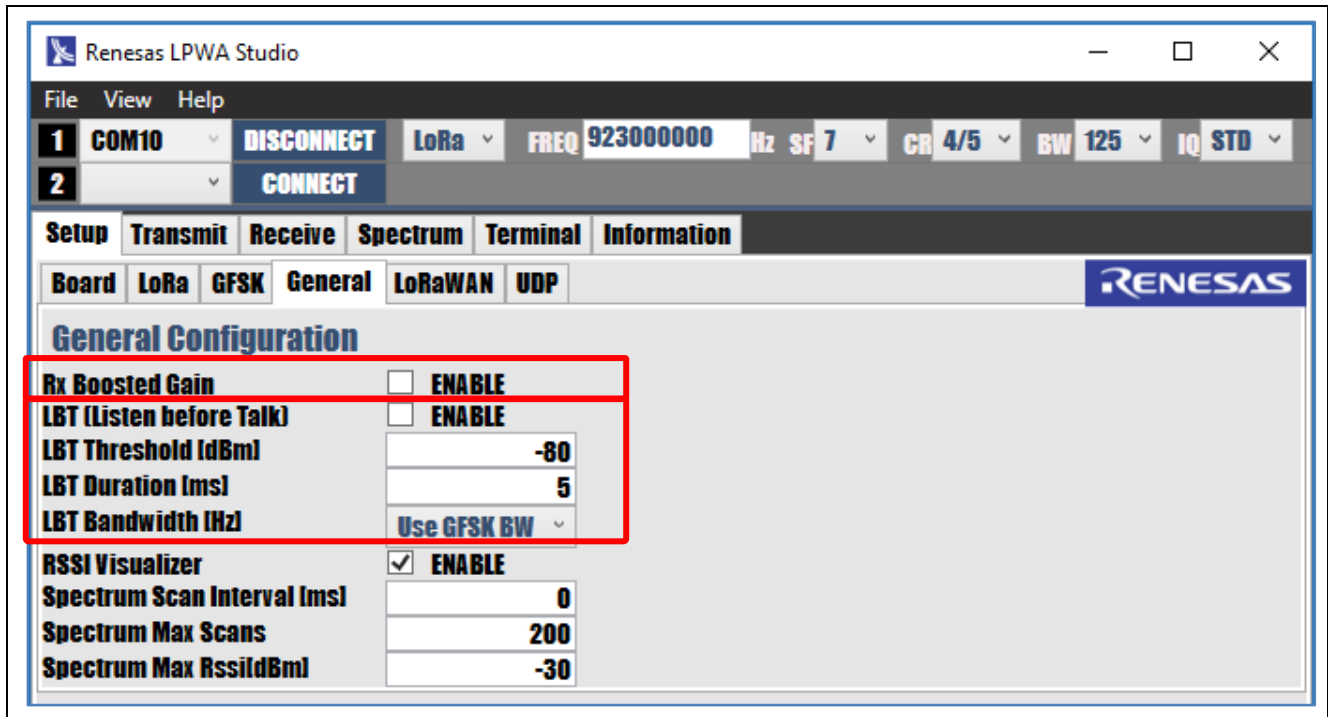


Figure 9. RX Boosted Gain and LBT Parameters Configuration

SNIFFER Spectrum

qq

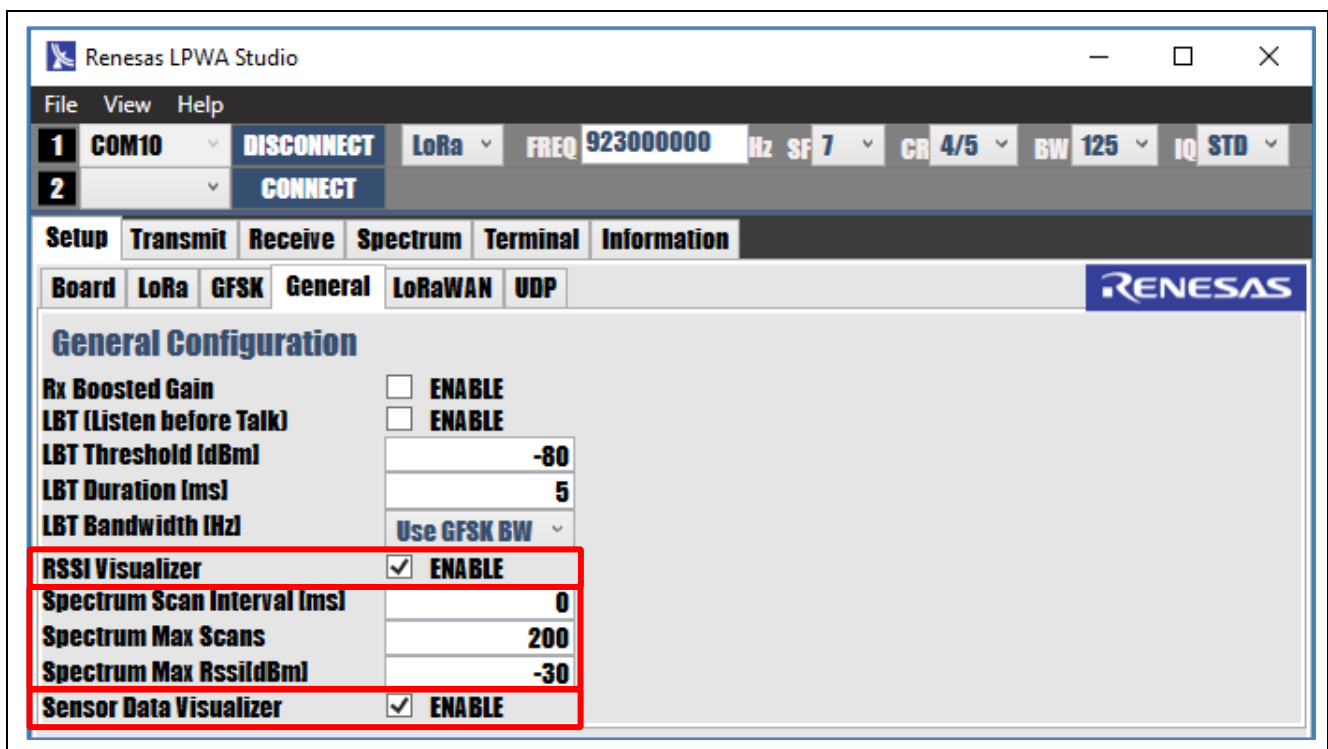


Figure 10. Sensor Data Visualizer

Menu File Reset All

4.4 Continuous Wave Transmission

- Transmit

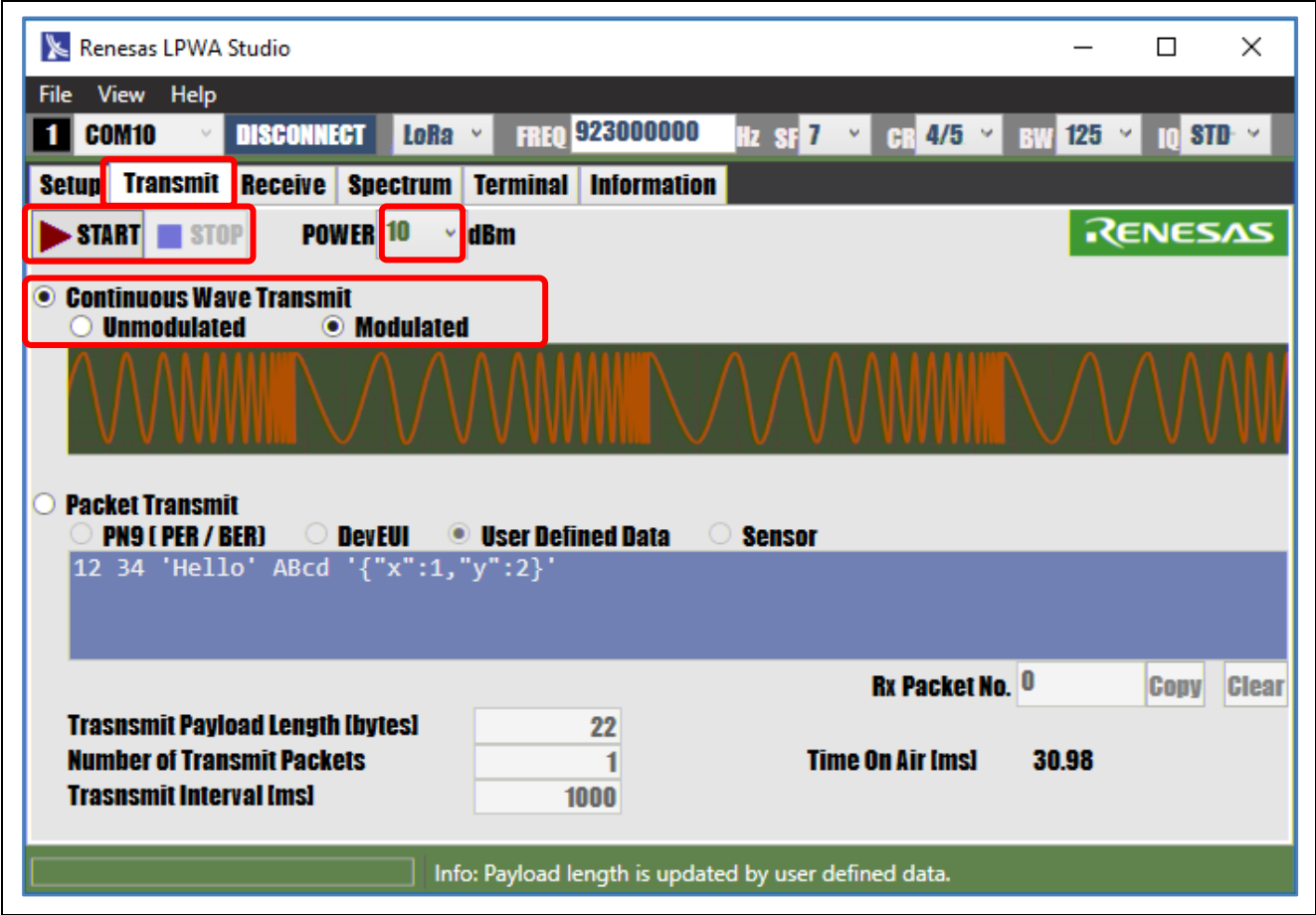


Figure 11. Continuous Wave Transmit Selection

LoRa TU TU GFSK

4.5 Packet Transmission

-

4.6 SNIFFER (Packet Reception)

Receive SNIFFER START STOP.

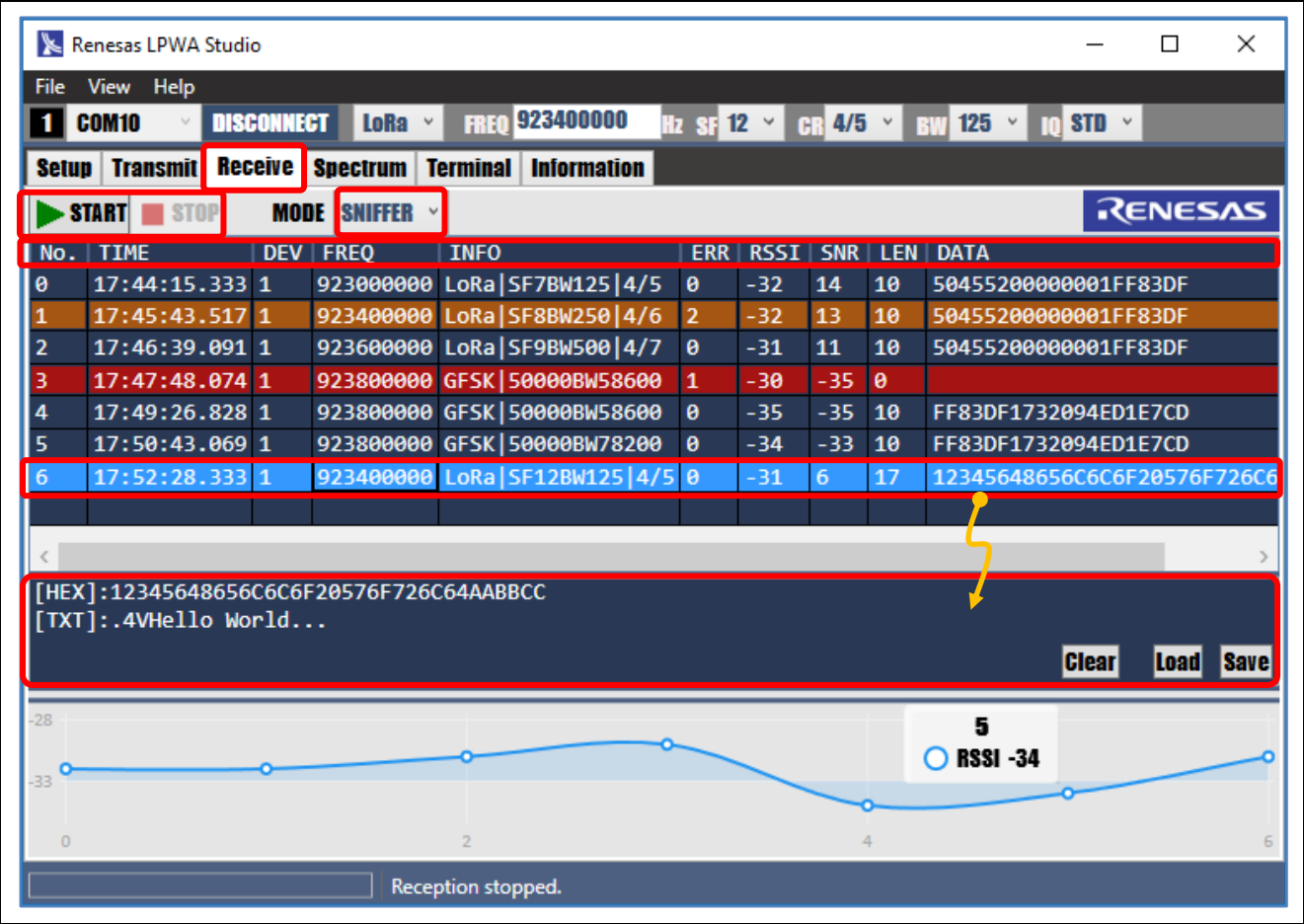


Figure 13. SNIFFER Mode

R SS S R S

Save Load

Setup

GFSK SNR

RS RS RS RS DEV

ERR

Setup

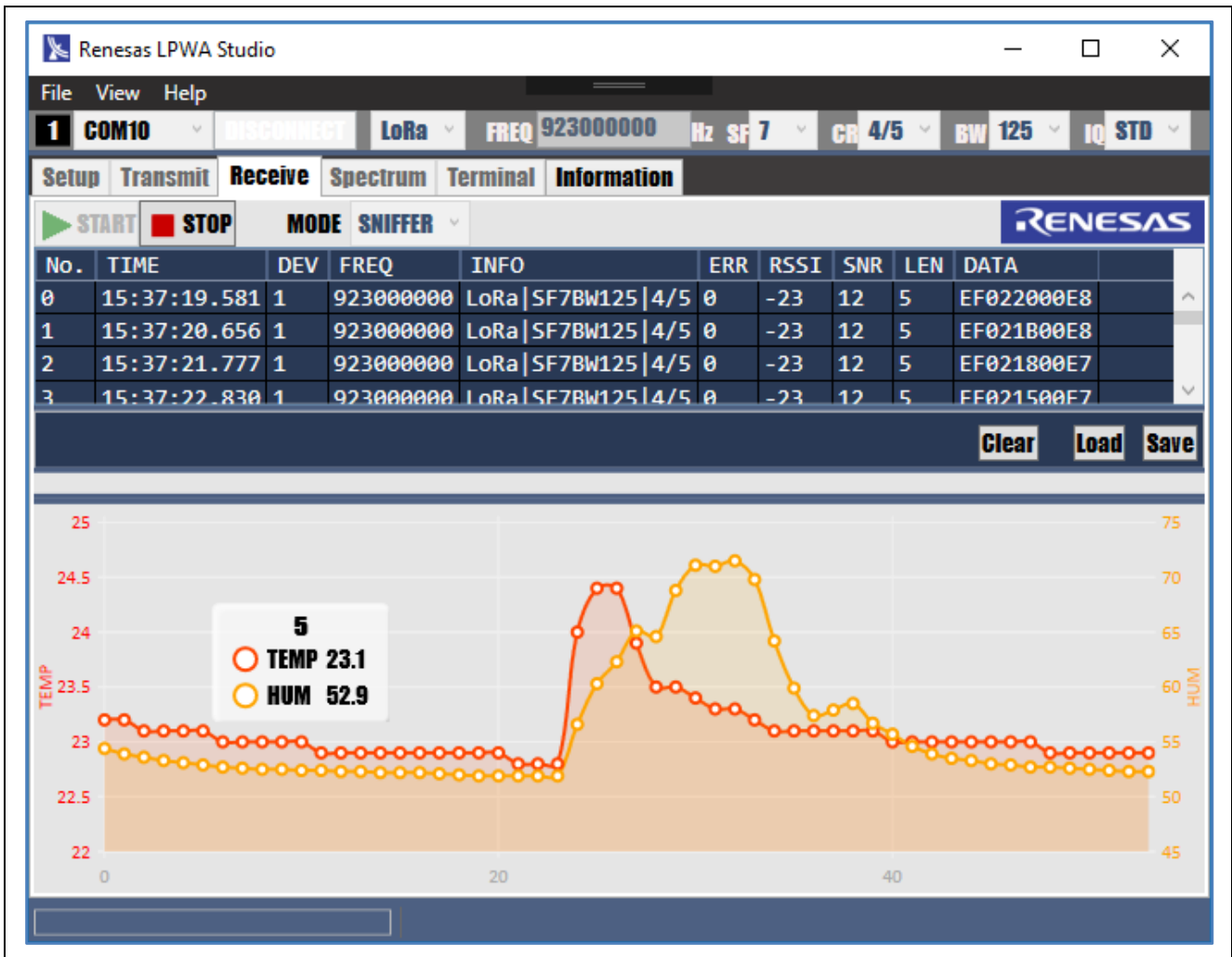


Figure 14. Sensor Data Visualization

4.7 PER/BER Test

Word Payload CRC Type Frame Header Type Setup Preamble Length Sync

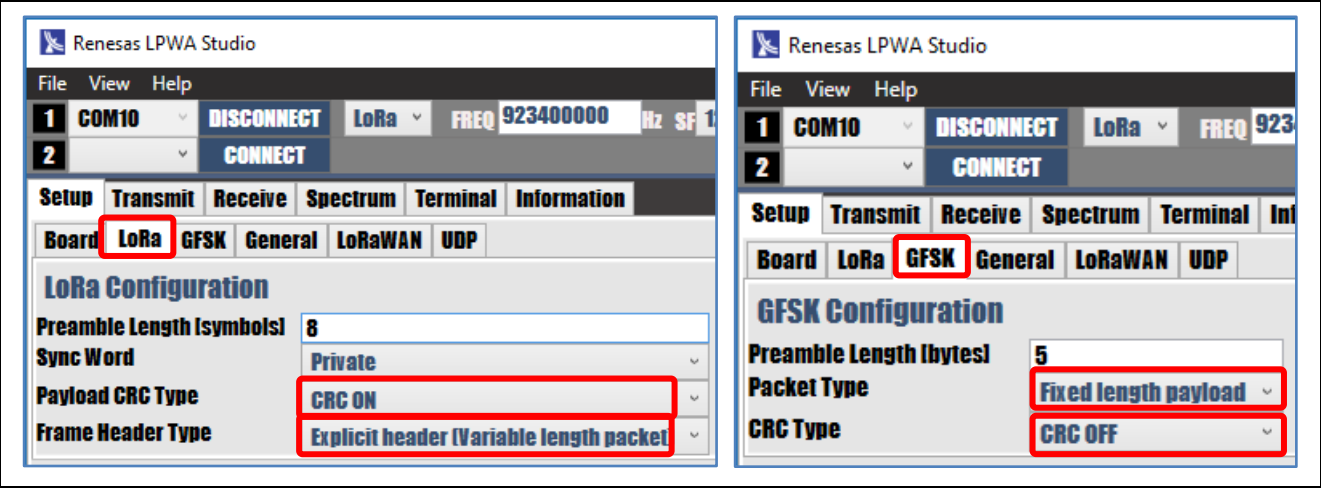


Figure 15. Setting LoRa and GFSK Configuration Parameters for PER/BER Testing

PN9 Data for PER/BER PER/BER

-
-

LoRa GFSK

GFSK Expected Data Length SNR

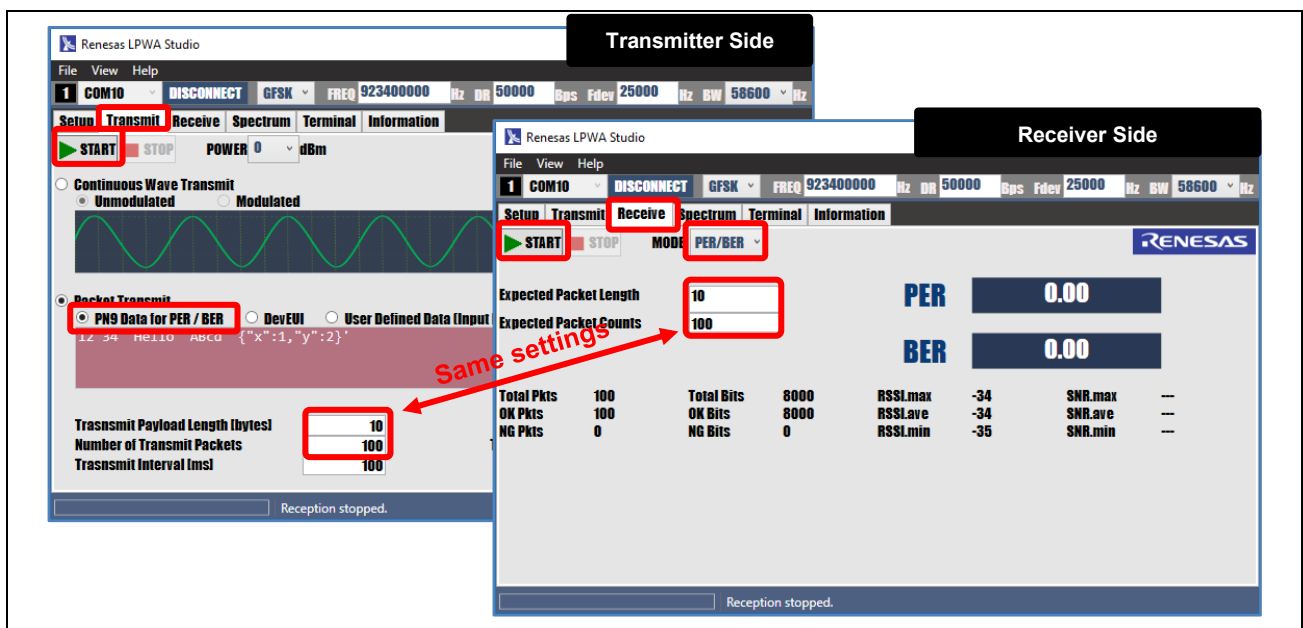


Figure 16. Transmitter and Receiver Side Settings

4.8 Spectrum Scan

Spectrum BEGIN/END/STEP Scan START R S

GFSK GFSK

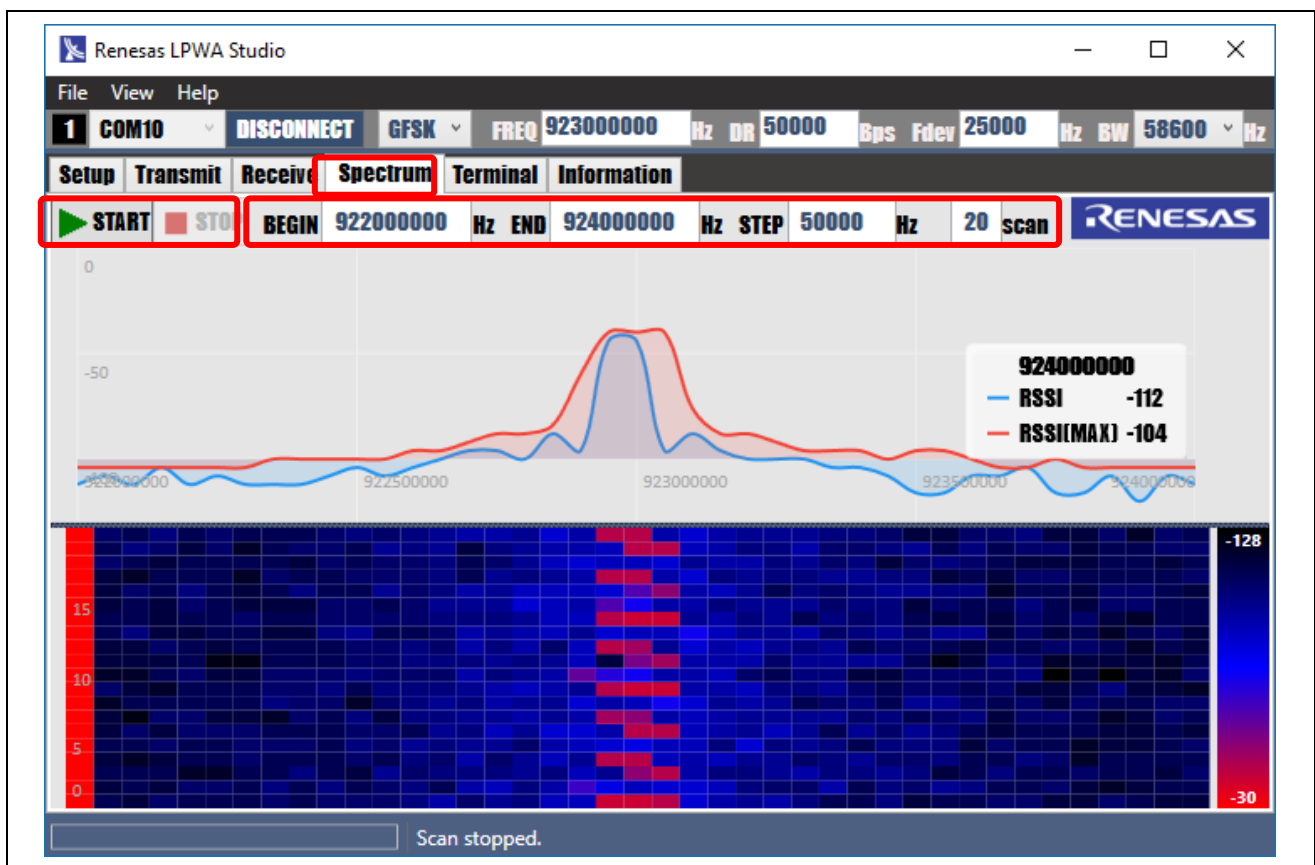


Figure 17. Spectrum Scan

Setup

The screenshot shows the Renesas LoRaWAN Setup interface. The 'General' tab is selected under the 'LoRa' section. The 'General Configuration' section contains the following settings:

Parameter	Value
Rx Boosted Gain	☐ ENABLE
LBT (Listen before Talk)	☐ ENABLE
LBT Threshold (dBm)	-80
LBT Duration (ms)	5
LBT Bandwidth (Hz)	Use GFSK BW
RSSI Visualizer	☒ ENABLE
Spectrum Scan Interval (ms)	0
Spectrum Max Scans	200
Spectrum Max Rssi (dBm)	-30

The last three rows of the table are highlighted with a red box.

Figure 18. Spectrum Scan Data
Max Rssi

4.9 Information

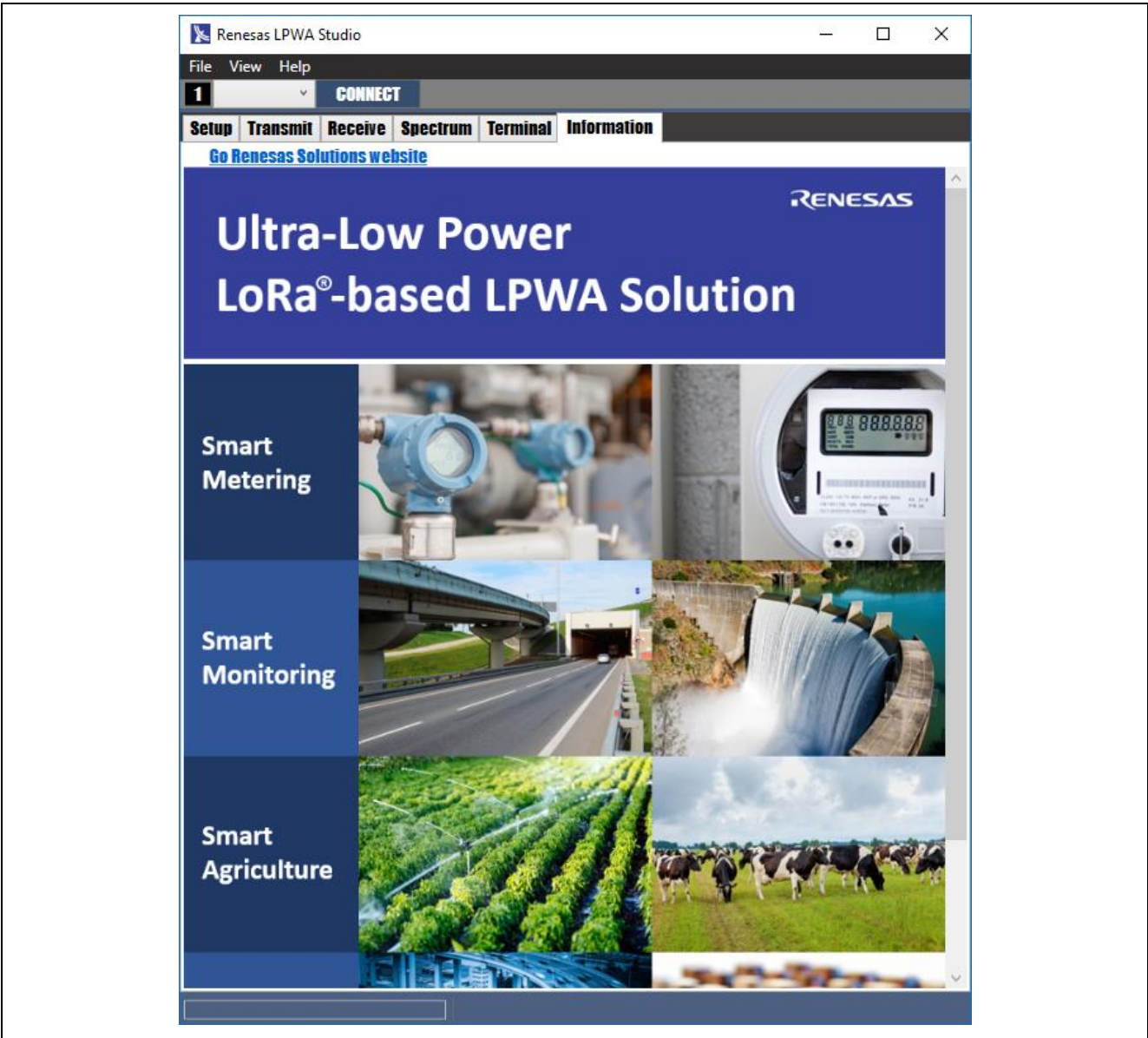


Figure 19. LoRa-based LPWA Solution Information

5. Using Renesas LPWA Studio [Advanced]

5.1 Terminal

Terminal R S
q Radio Evaluation Program Application Note

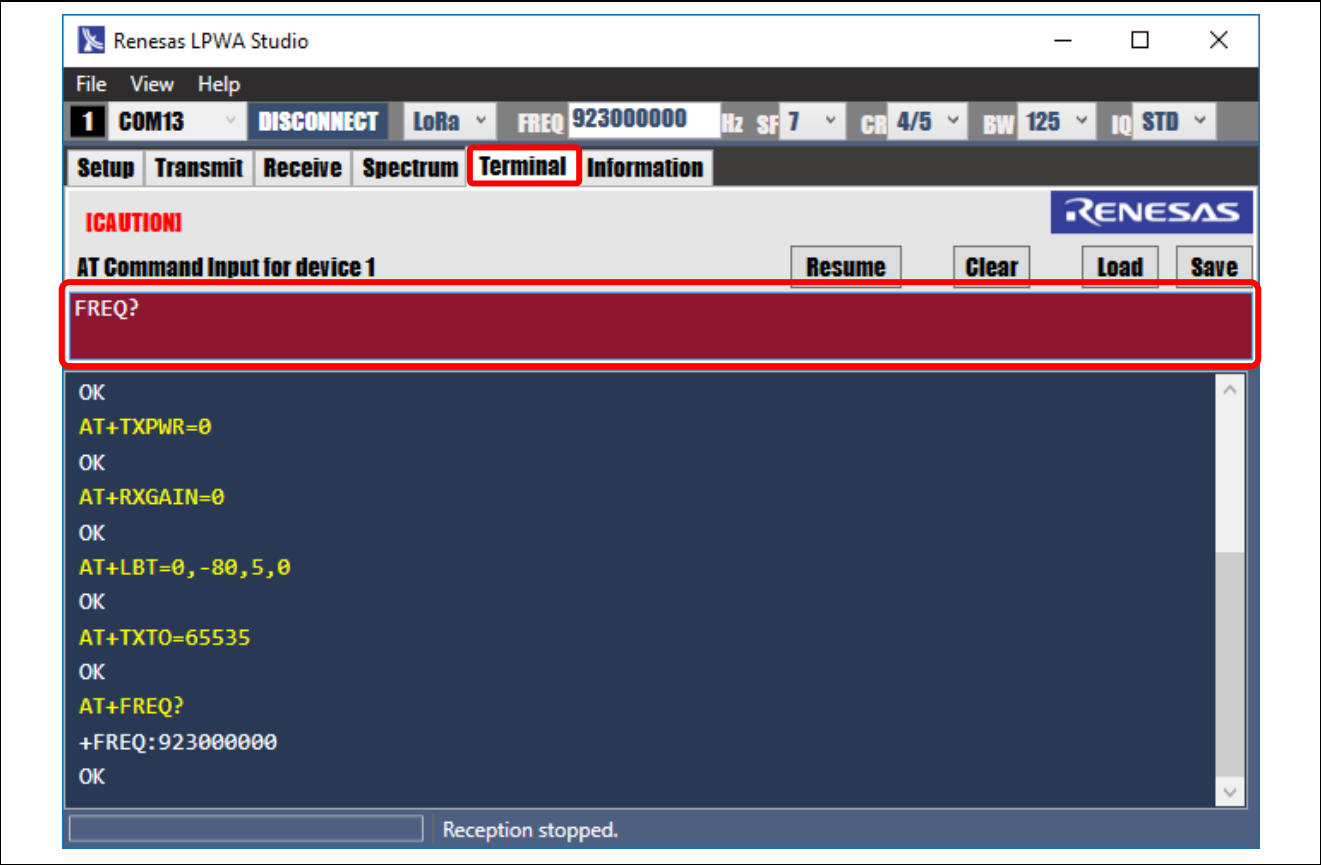


Figure 20. Terminal View

Resume

Save

Load Drag & Drop

```
// Sample Script
AT+MODEM=1           // LoRa Modulation
AT+FREQ=923400000    // Frequency 923.4MHz
AT+TXPWR=10          // Transmission Power 10 dBm
AT+TXCP              // Start continuous modulated wave transmission
@wait 5000           // wait 5000 ms
```

Figure 21. Script Example

5.2 Capturing the LoRaWAN Packets

Case 1

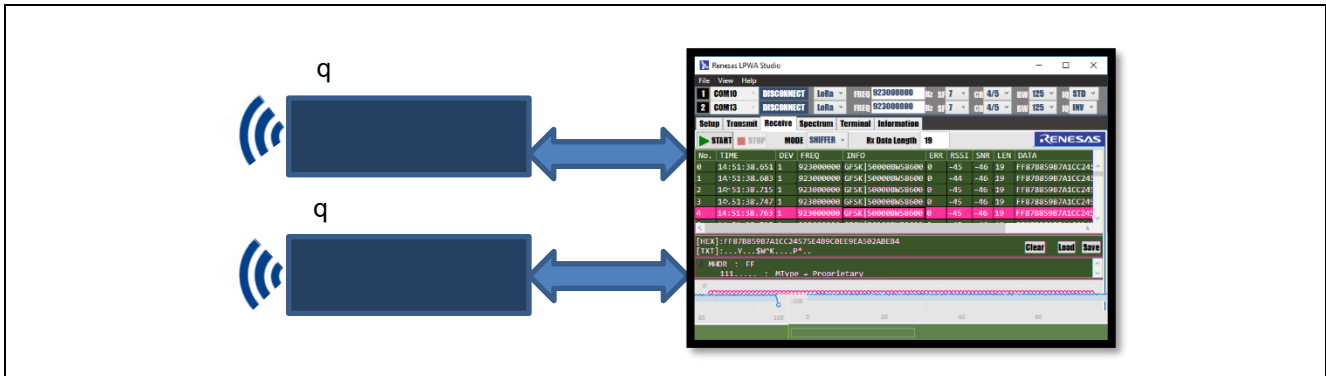


Figure 22. Two Target Boards with Serial USB (Single Channel Capture)

Case 2

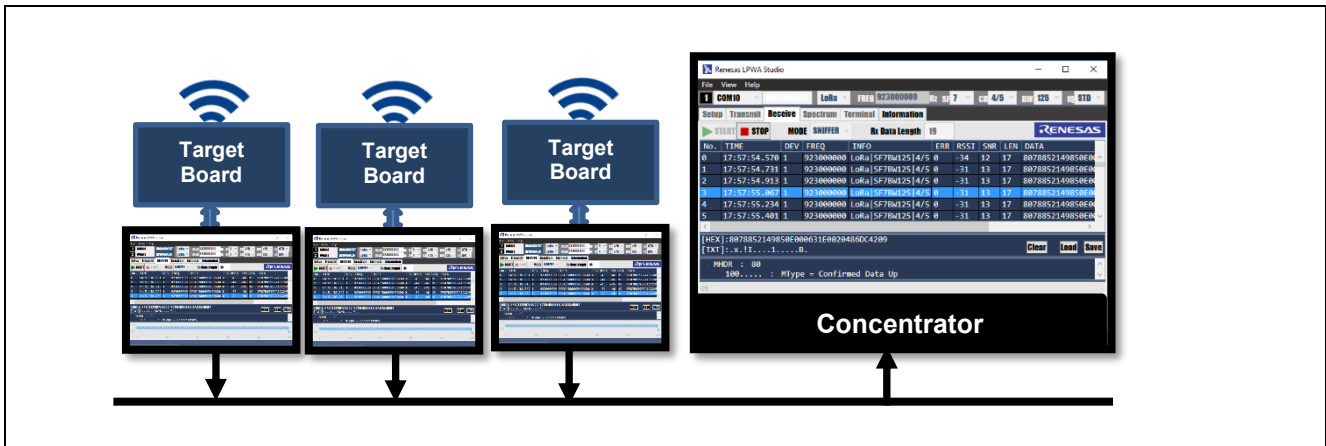


Figure 23. Multiple Target Boards (Multi Channel Capture)

Case 3

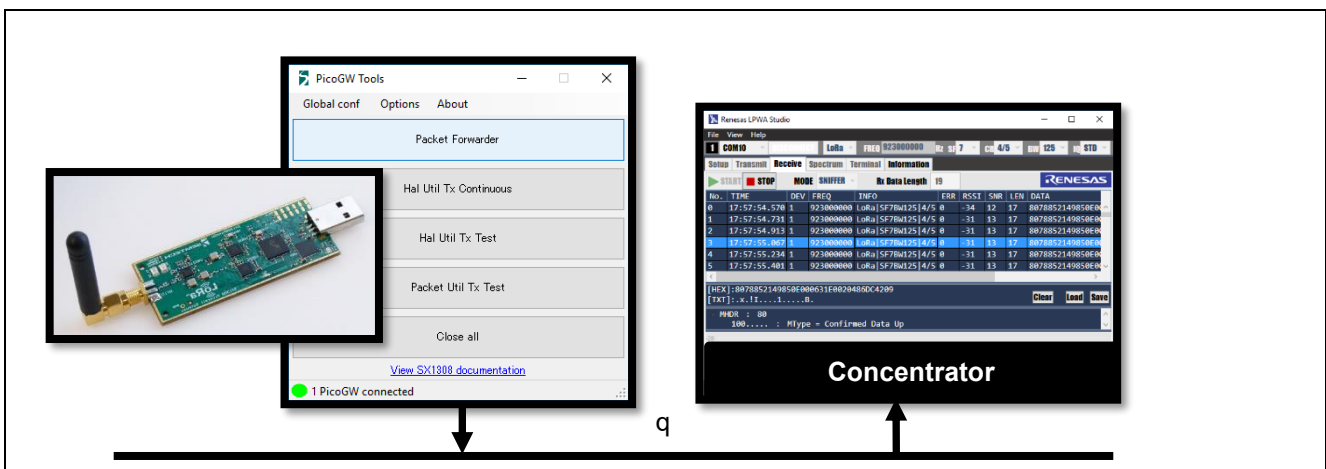


Figure 24. Semtech Packet Forwarder (Maximum of 8 Channels Capture)

Case 4

5.2.1 Case 1 Dual Target Board

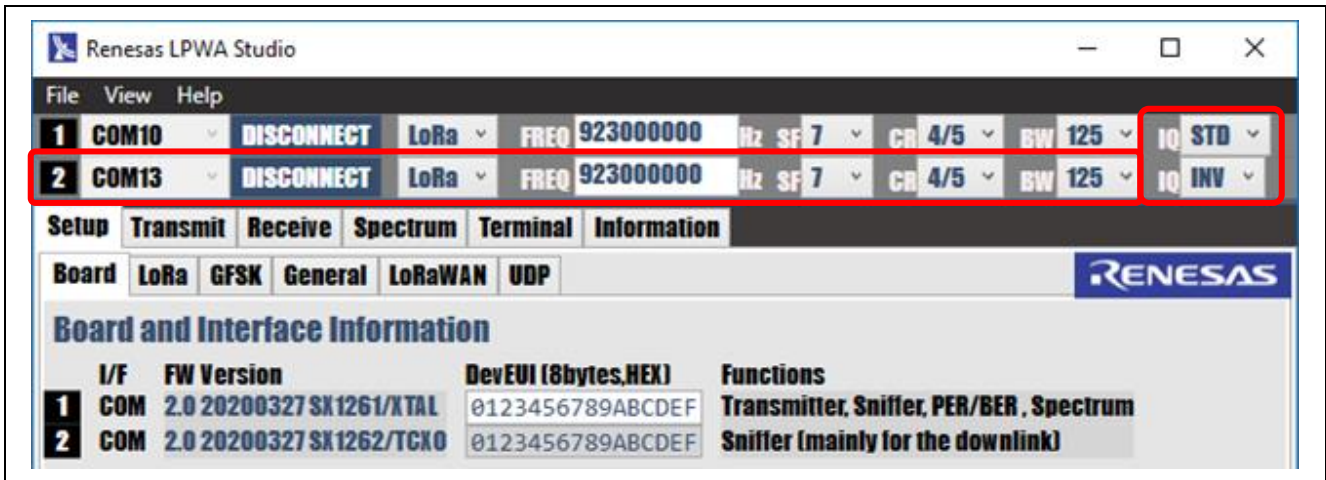


Figure 25. Two Target Boards Case 1

5.2.2 Case 2 Multiple Renesas LPWA Studio

IMPORTANT

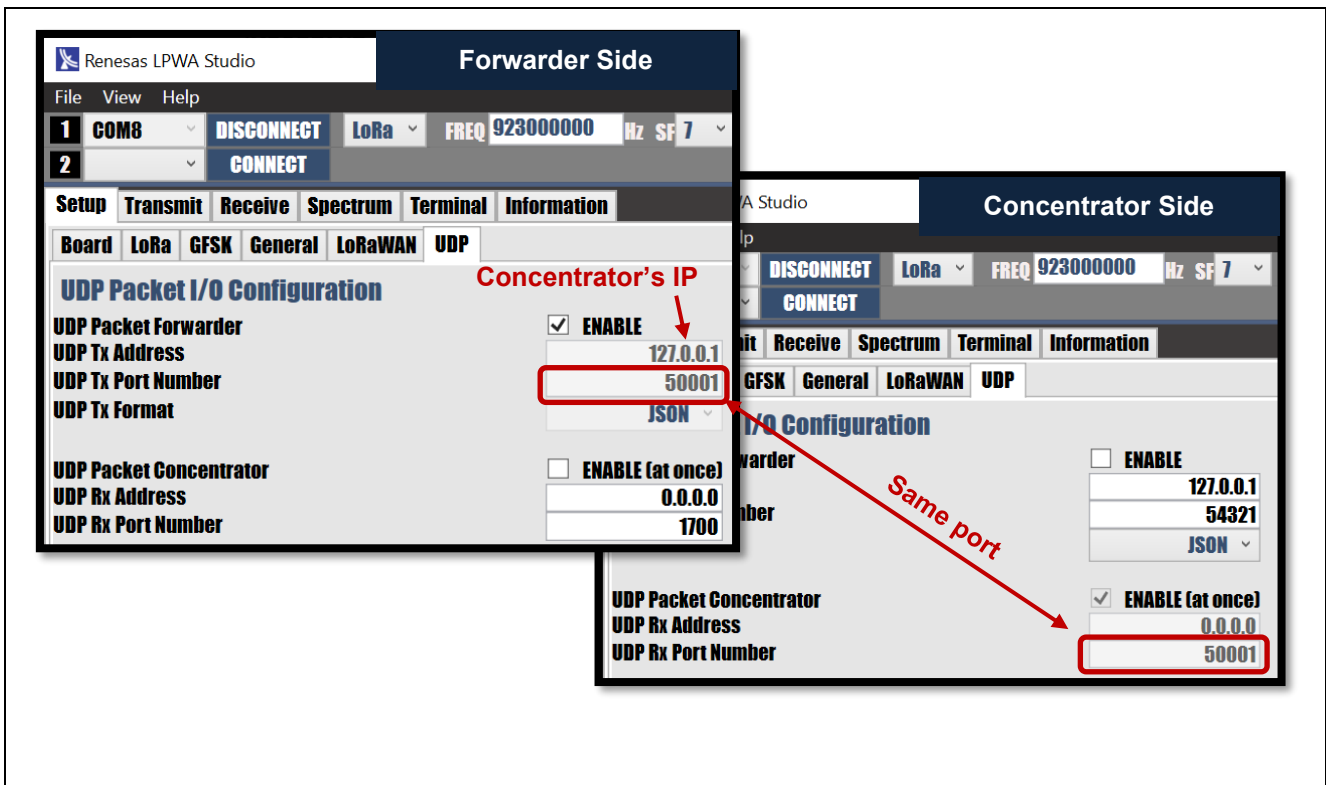


Figure 26. Multiple Target Boards Case 2

5.2.3 Case 3 Semtech Packet Forwarder

q

q qq
q

q q U

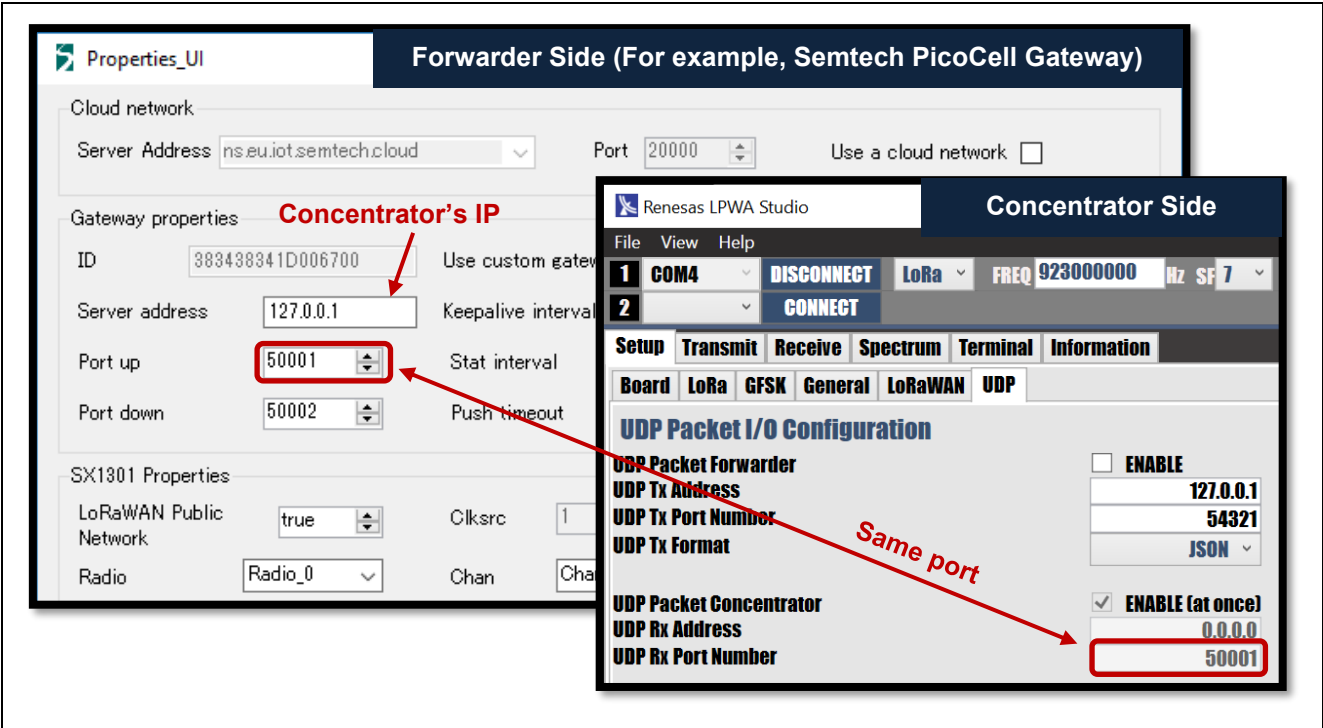


Figure 27. Semtech Packer Forwarder Case 3

5.3 Built-in LoRaWAN Decoder

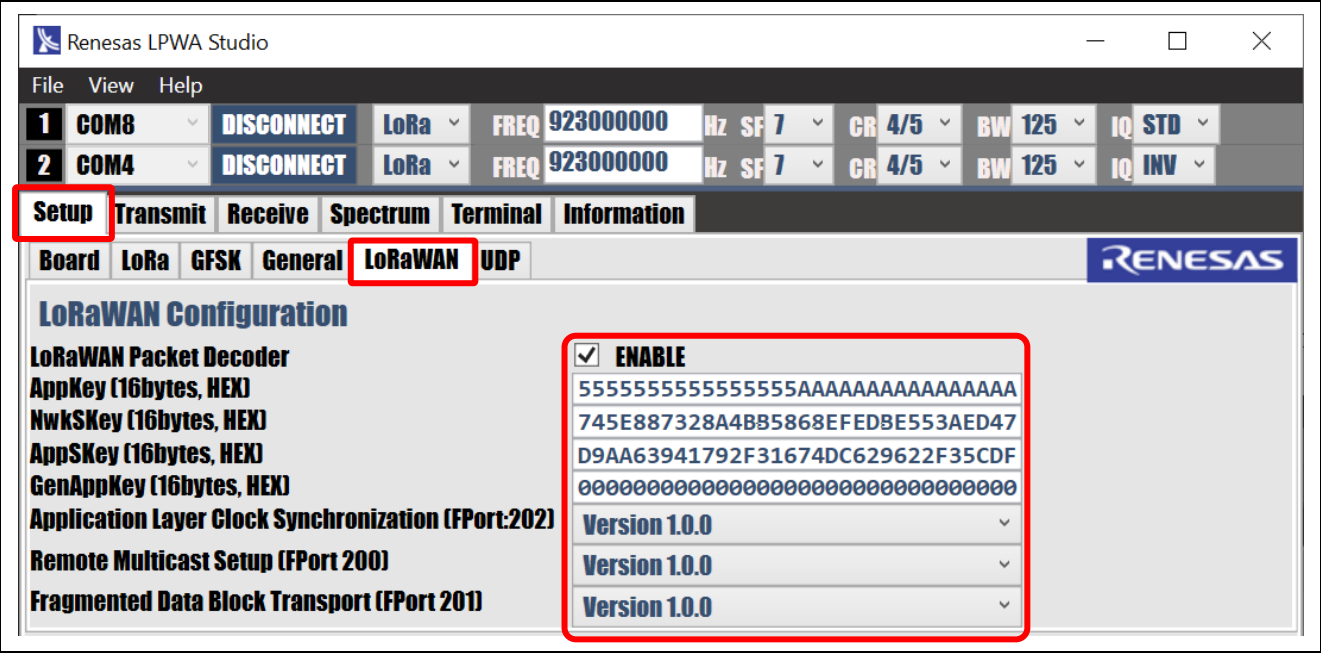
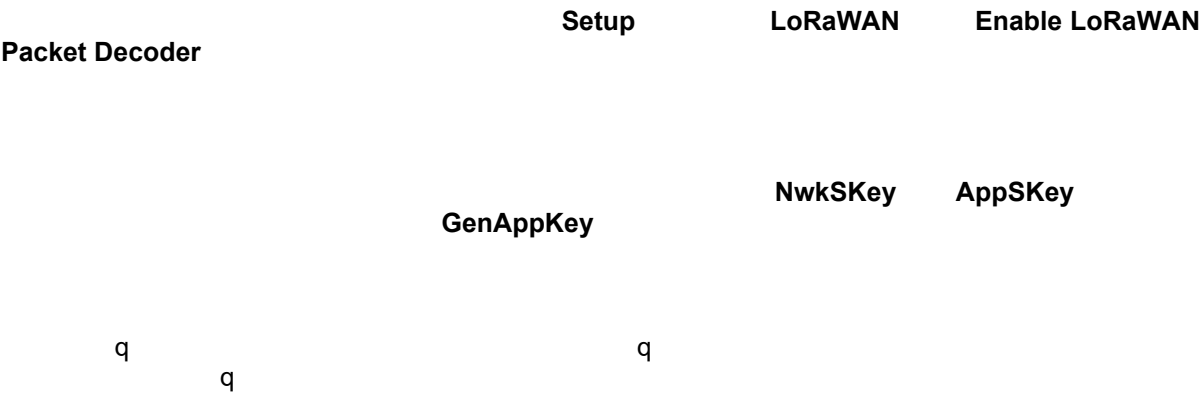


Figure 28. Enabling Built-in LoRaWAN Decoder



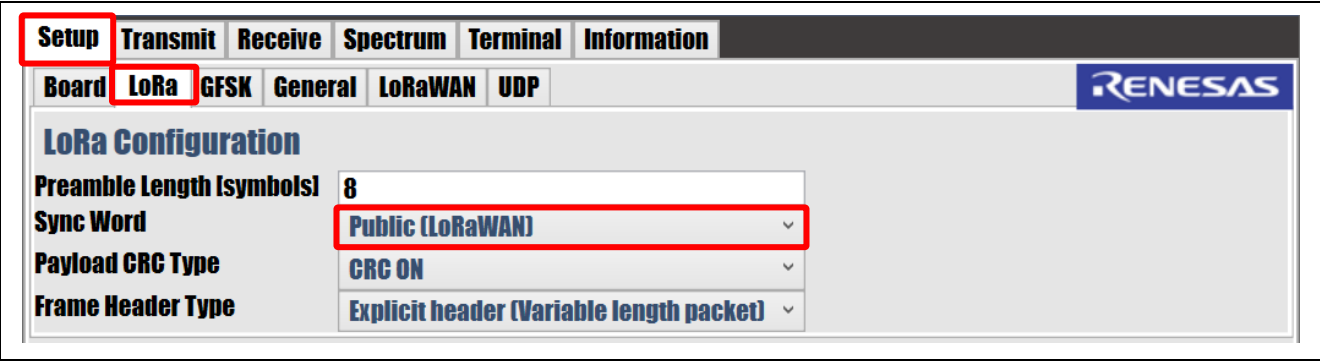


Figure 29. Setting LoRa Sync Word to Public

Public

Receive SNIFFER Start

Clip

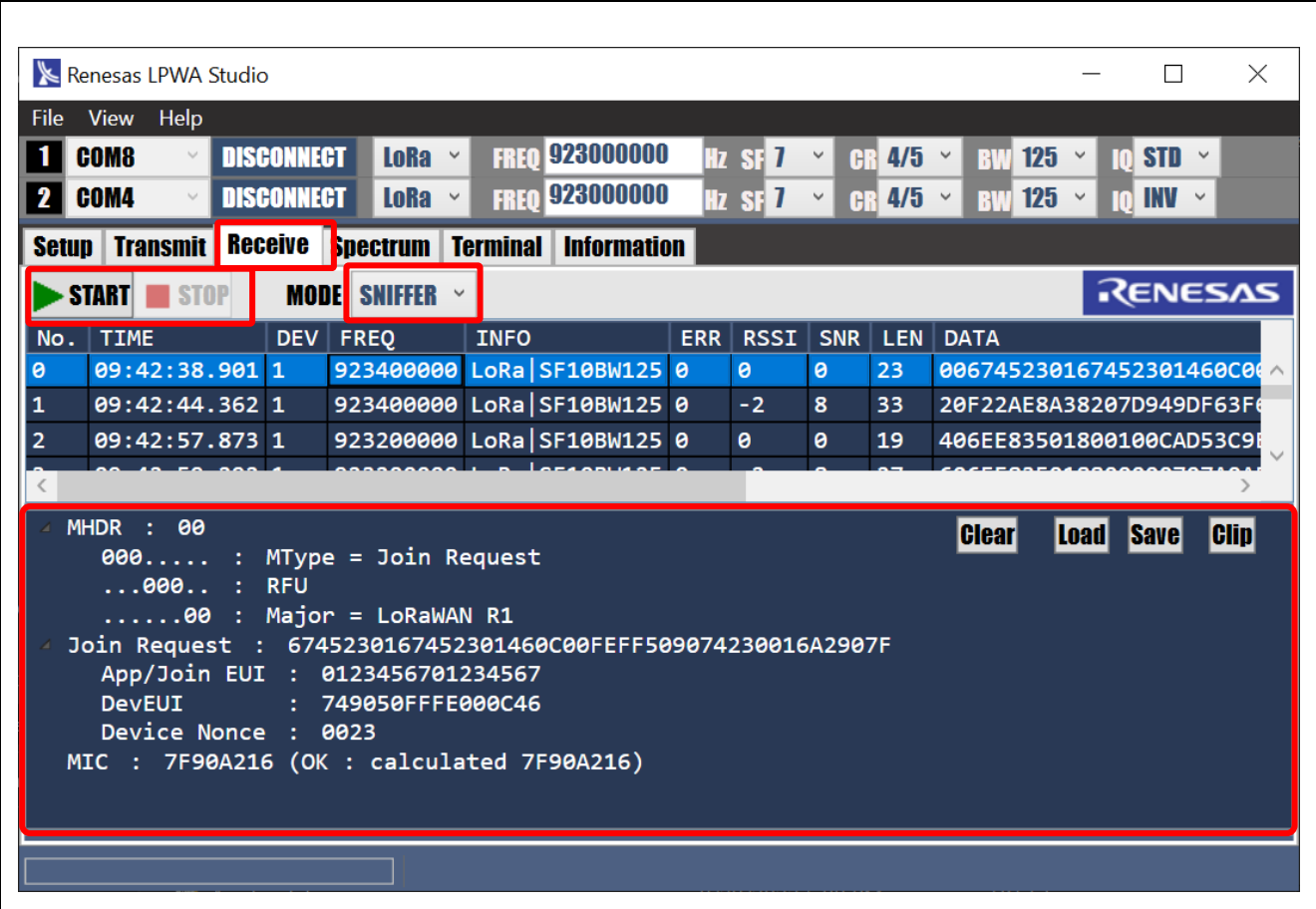


Figure 30. Displaying the Received LoRaWAN Packet

5.4 Packet Forwarder

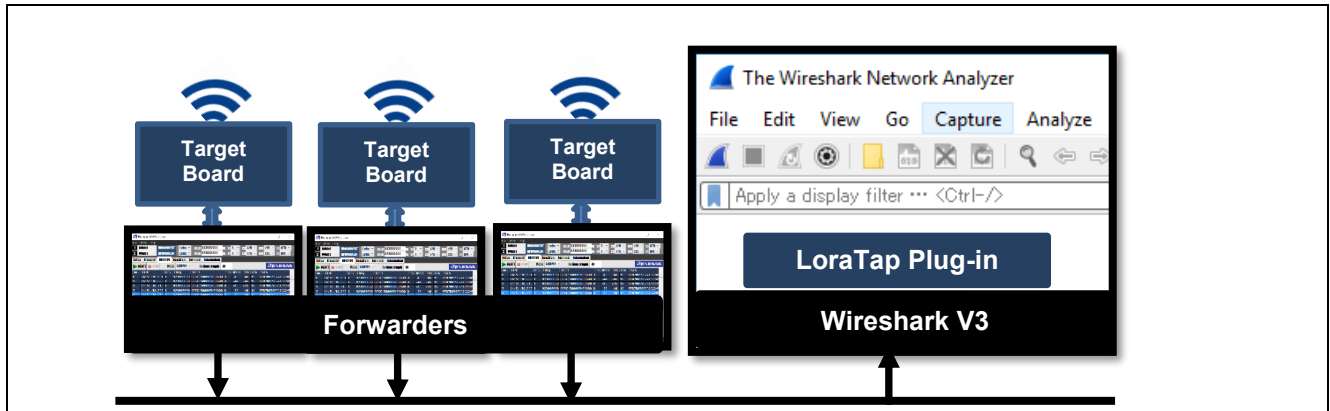


Figure 31. Forwarding the Received LoRaWAN Packets (Wireshark)

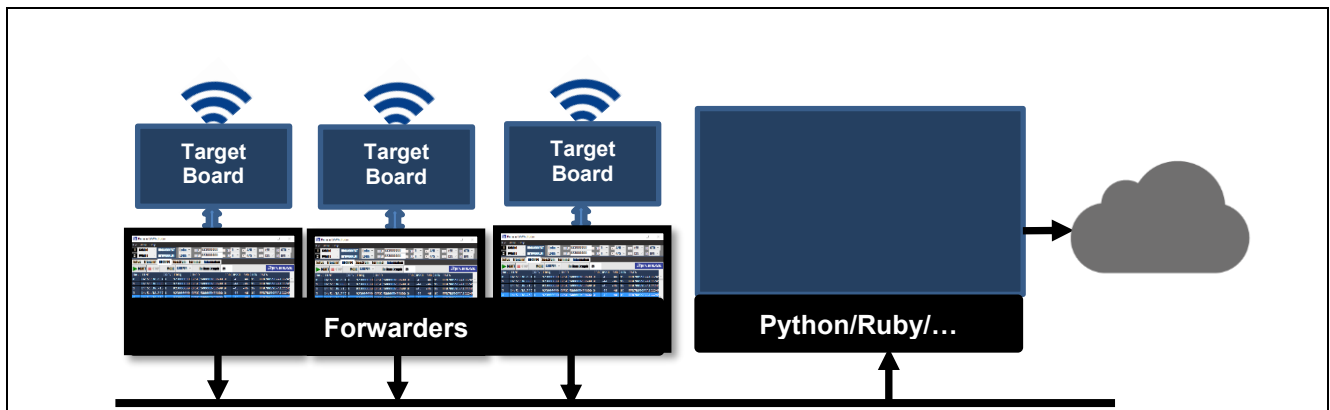


Figure 32. Forwarding the Received LoRaWAN Packets (External Applications)

5.4.1 Case 1: Wireshark

Setup UDP UDP Packet Forwarder

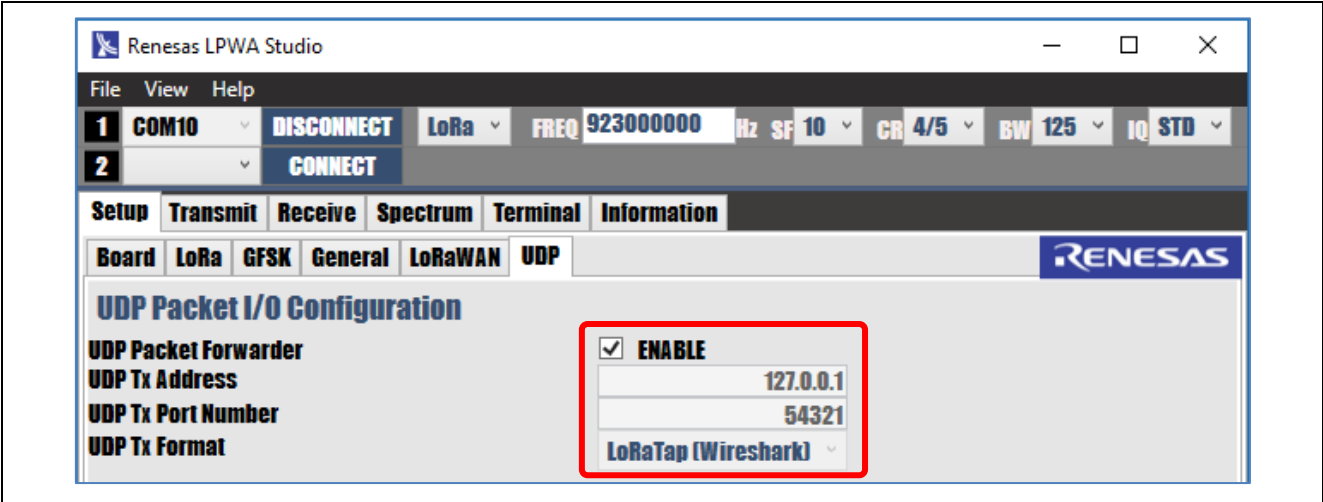


Figure 33. Configuring the UDP Packet Forwarder for Wireshark

Receive SNIFFER Start
udp port 54321 Adapter for loopback traffic capture

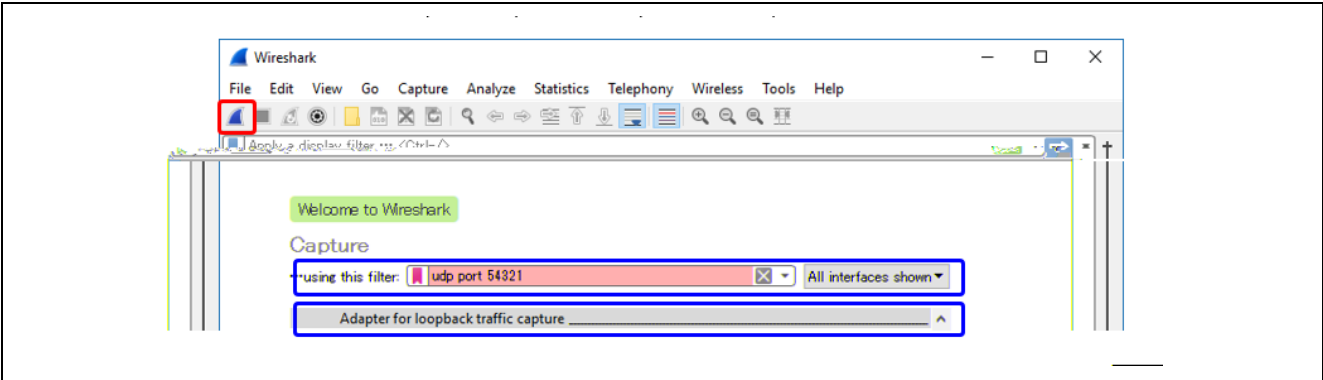


Figure 34. Using Wireshark for Packet Forwarding

5.4.2 Case 2: External Programs

Setup

UDP

UDP Packet Forwarder

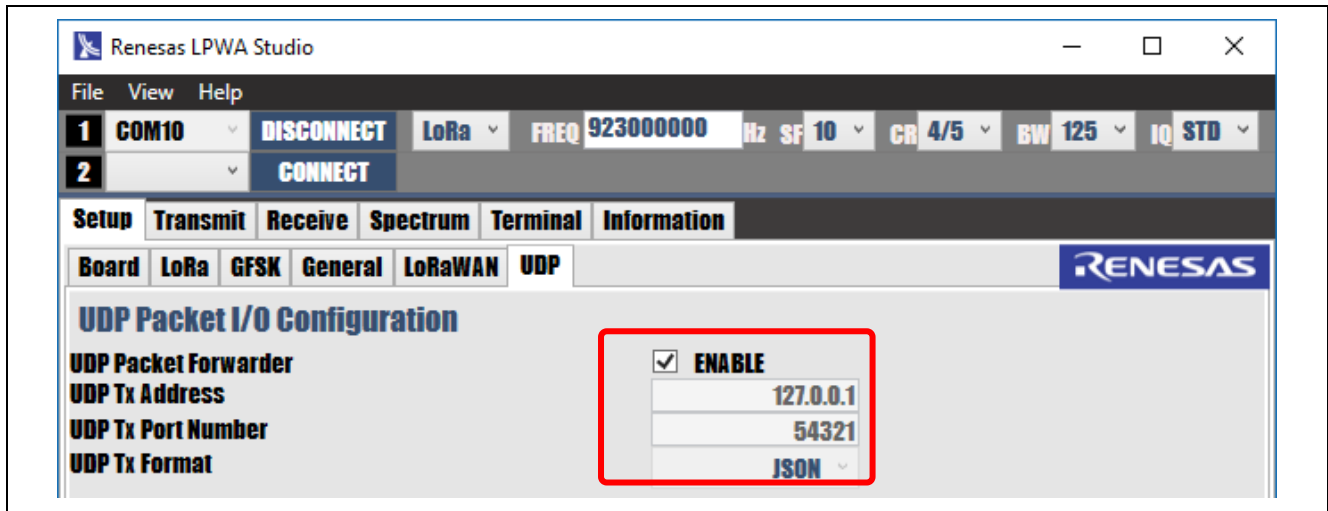


Figure 35. Configuring the UDP Packet Forwarder for External Programs

Receive

SNIFFER

Start

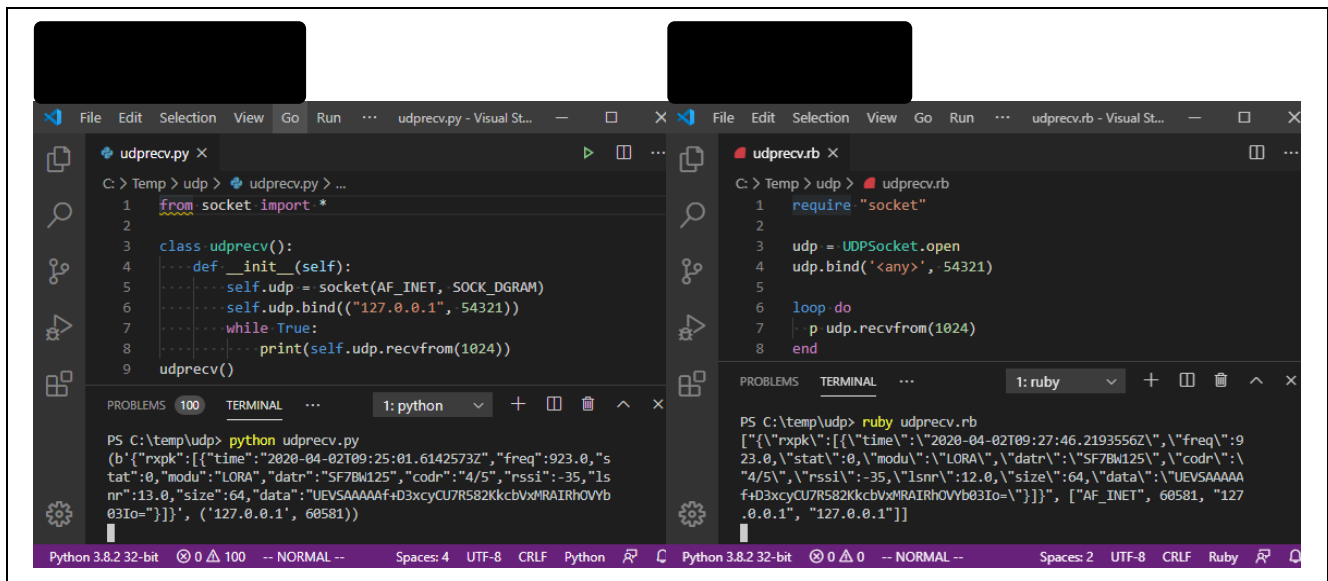


Figure 36. Example Scripts to Receive UDP Packets

- InfluxDB 1.8.x Grafana 7.1.x

-

-

-

-

home

8086

UDP Packet forwarder
sensor node

FROM sensor_demo WHERE deveui = lowest byte of the DevEUI
SELECT temperature or humidity

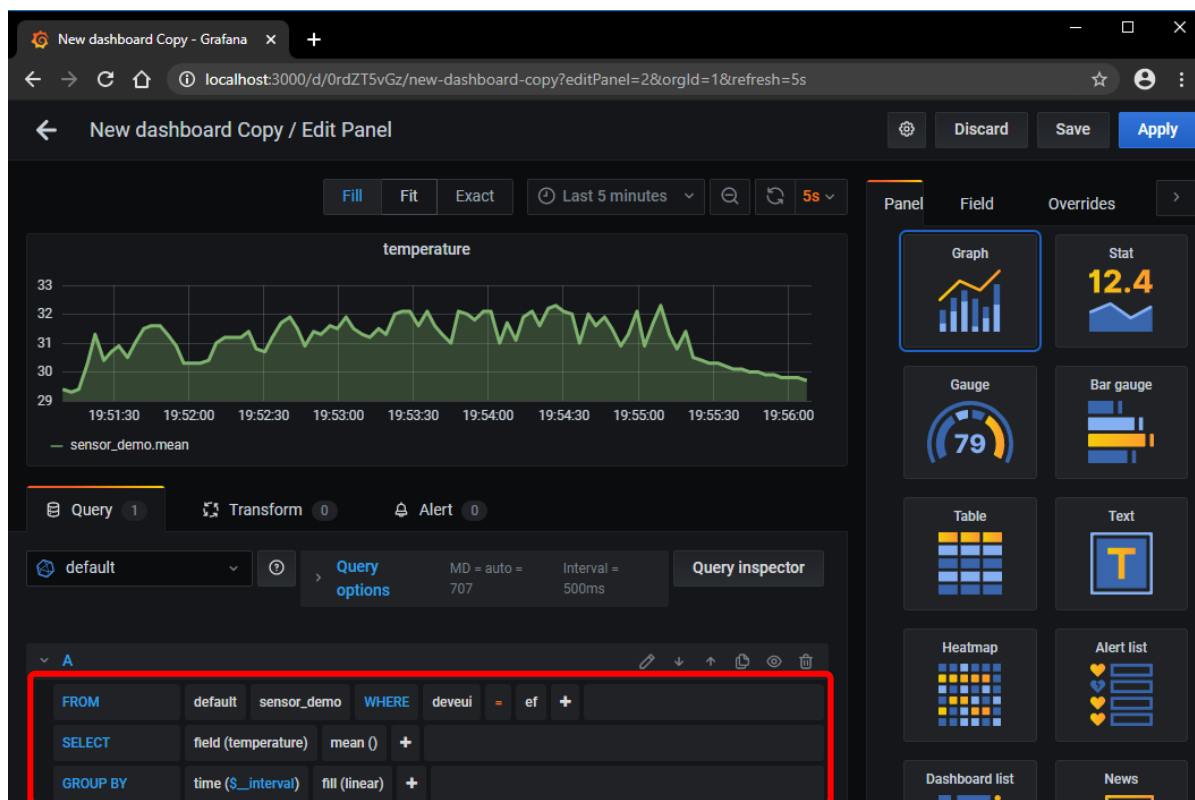


Figure 37. Sensor Data Visualization Demo with UDP Packet Forwarding Function

			q
			q

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